

LAPORAN

TUGAS MATA KULIAH
STRUKTUR DATA
SEMESTER II



(Tugas 2 Program Bilangan Struktur Data)

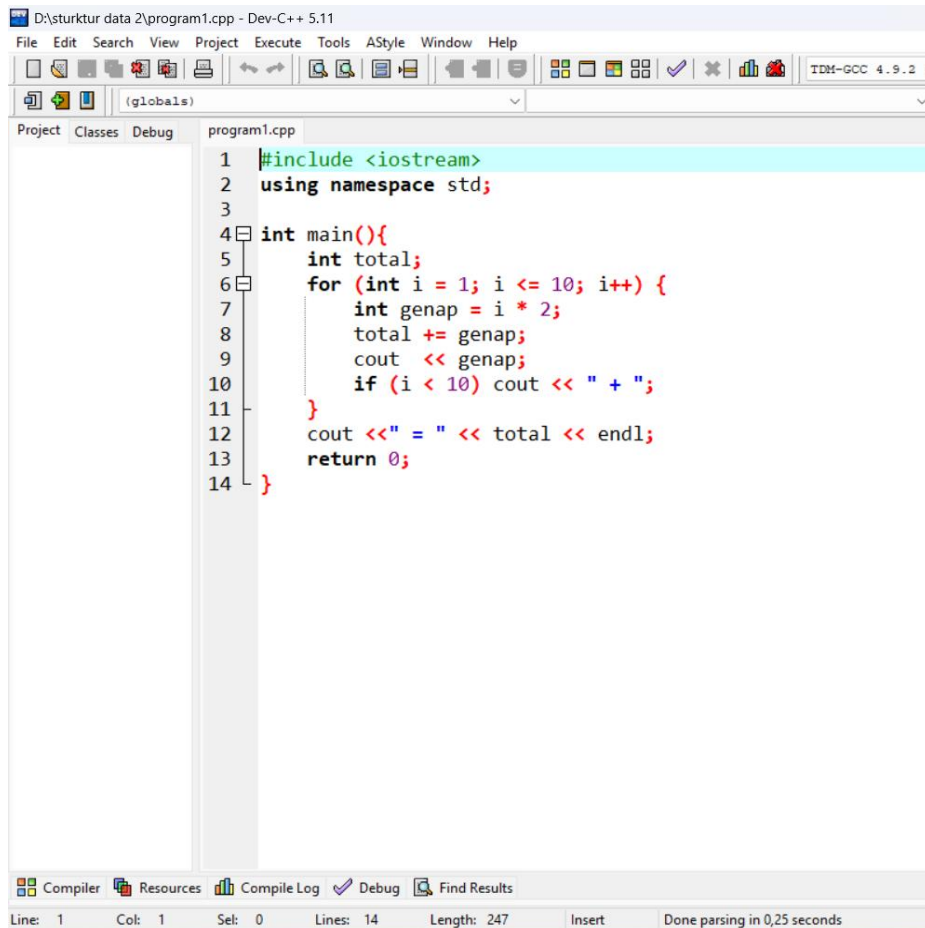
Oleh:
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PROGRAM STUDI SISTEM INFORMASI
INSTITUT TEKNOLOGI MOJOSARI
NGANJUK
2025

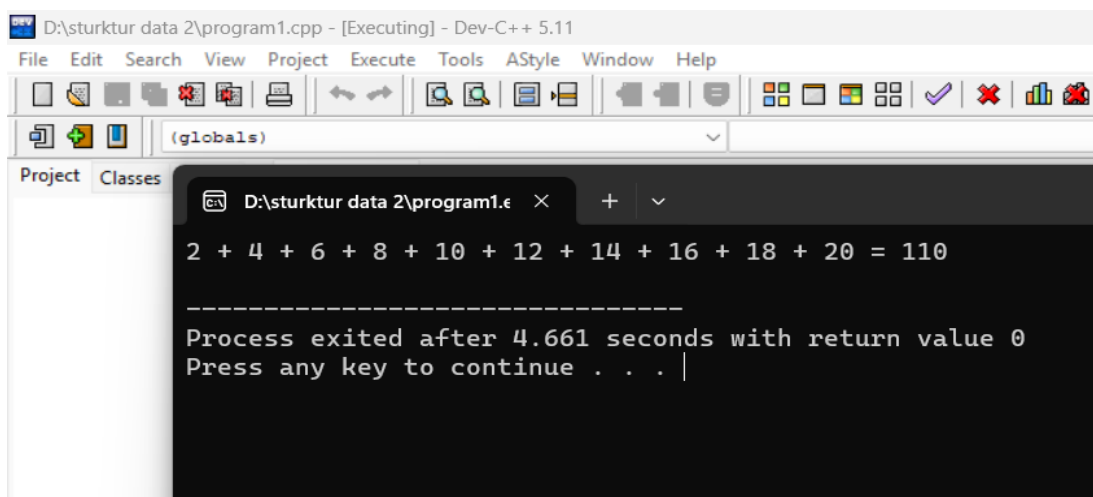
1. Soal Pertama

bilangan genap dengan hasilnya : $2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20 = 110$:

Jawaban:



```
1 #include <iostream>
2 using namespace std;
3
4 int main(){
5     int total;
6     for (int i = 1; i <= 10; i++) {
7         int genap = i * 2;
8         total += genap;
9         cout << genap;
10        if (i < 10) cout << " + ";
11    }
12    cout << " = " << total << endl;
13    return 0;
14 }
```

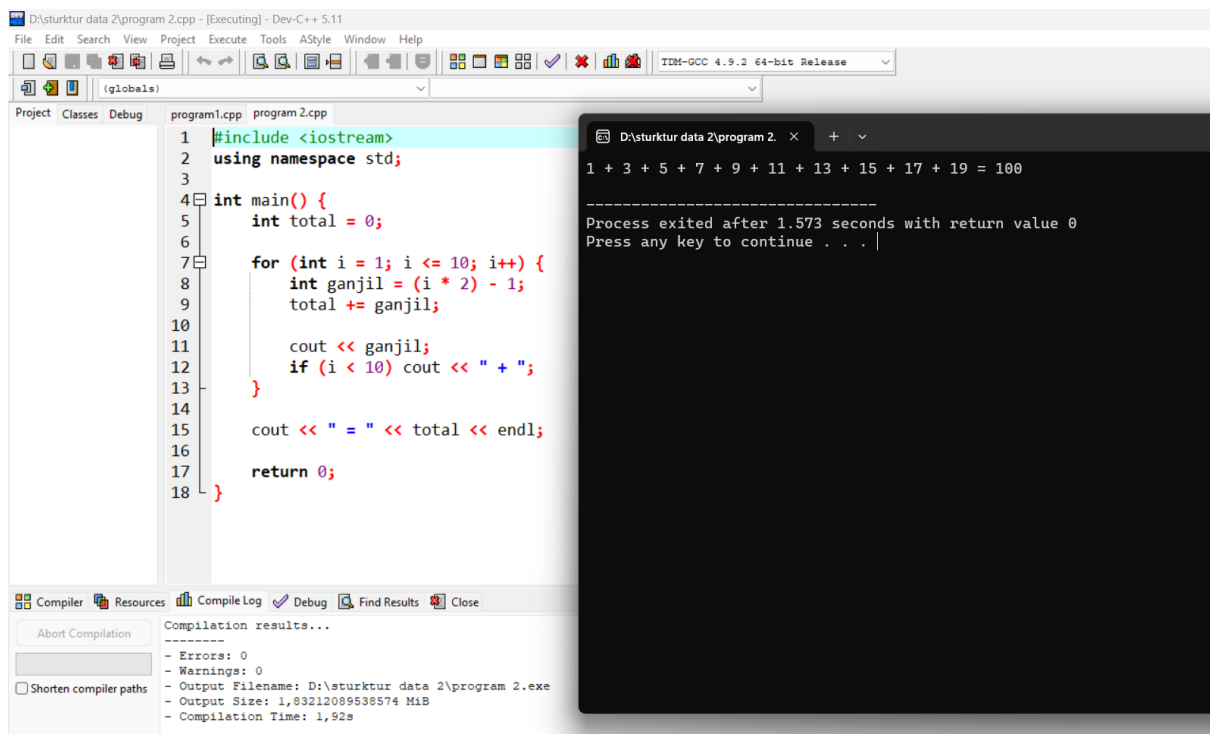


```
D:\stuktur data 2\program1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
D:\stuktur data 2\program1.cpp
2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20 = 110
-----
Process exited after 4.661 seconds with return value 0
Press any key to continue . . .
```

2. Soal Ke dua

menghitung 10 deret bilangan ganjil dengan hasilnya : $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 100$

Jawaban:



The screenshot displays a C++ development environment. The main editor window shows the source code for 'program2.cpp', which calculates the sum of the first 10 odd numbers. The code uses a loop to generate odd numbers and accumulate their sum, then prints the result. A secondary window shows the program's output, which matches the expected result of 100. At the bottom, a 'Compiler' window provides details about the compilation process, including the output file name and compilation time.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int total = 0;
6
7     for (int i = 1; i <= 10; i++) {
8         int ganjil = (i * 2) - 1;
9         total += ganjil;
10
11         cout << ganjil;
12         if (i < 10) cout << " + ";
13     }
14
15     cout << " = " << total << endl;
16
17     return 0;
18 }
```

Output:

```
1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 100
-----
Process exited after 1.573 seconds with return value 0
Press any key to continue . . .
```

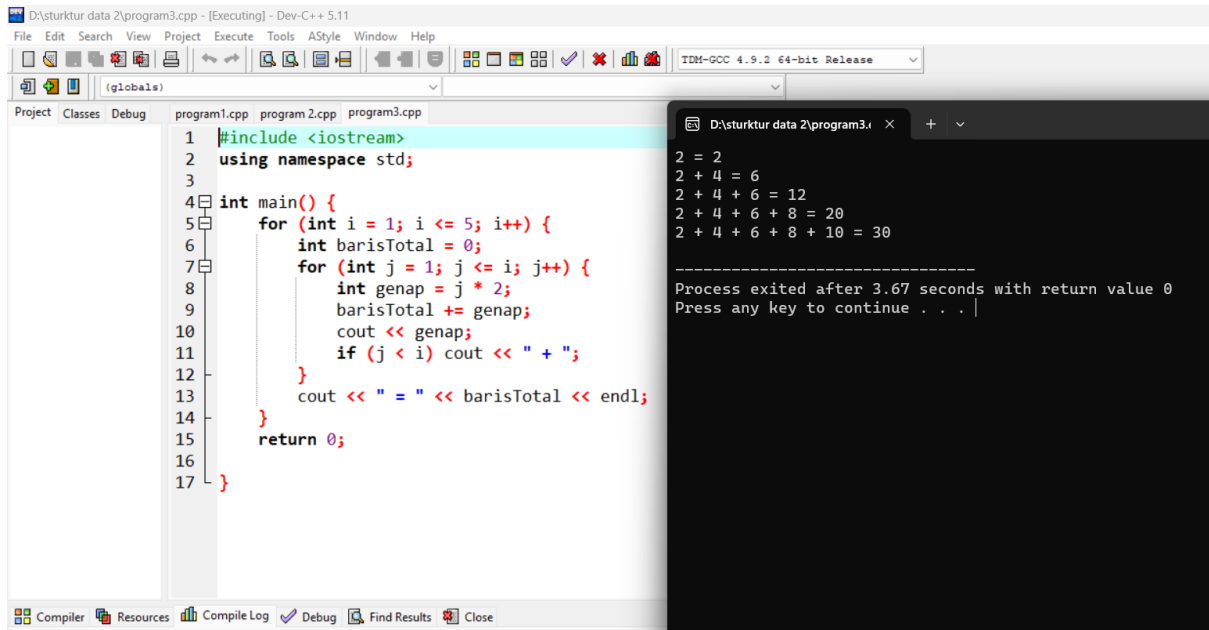
Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: D:\struktur data 2\program 2.exe
- Output Size: 1,83212089538574 MiB
- Compilation Time: 1,92s

3. Soal Ke tiga

Buatlah program untuk menghitung penjumlahan deret bilangan genap membentuk segitiga siku dengan hasilnya :

JAWABAN:



```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     for (int i = 1; i <= 5; i++) {
6         int barisTotal = 0;
7         for (int j = 1; j <= i; j++) {
8             int genap = j * 2;
9             barisTotal += genap;
10            cout << genap;
11            if (j < i) cout << " + ";
12        }
13        cout << " = " << barisTotal << endl;
14    }
15    return 0;
16 }
```

Output:

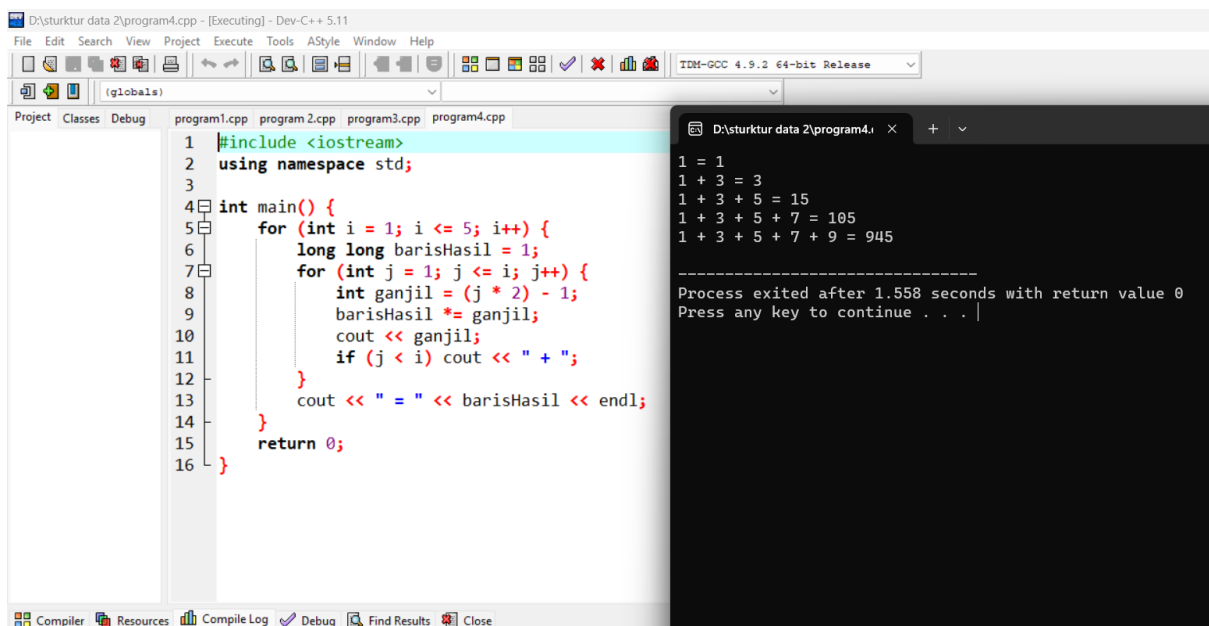
```
2 = 2
2 + 4 = 6
2 + 4 + 6 = 12
2 + 4 + 6 + 8 = 20
2 + 4 + 6 + 8 + 10 = 30

-----
Process exited after 3.67 seconds with return value 0
Press any key to continue . . .
```

4. Soal ke Empat

Buatlah program untuk menghitung perkalian deret bilangan ganjil membentuk segitiga siku dengan hasilnya :

JAWABAN:



```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     for (int i = 1; i <= 5; i++) {
6         long long barisHasil = 1;
7         for (int j = 1; j <= i; j++) {
8             int ganjil = (j * 2) - 1;
9             barisHasil *= ganjil;
10            cout << ganjil;
11            if (j < i) cout << " * ";
12        }
13        cout << " = " << barisHasil << endl;
14    }
15    return 0;
16 }
```

Output:

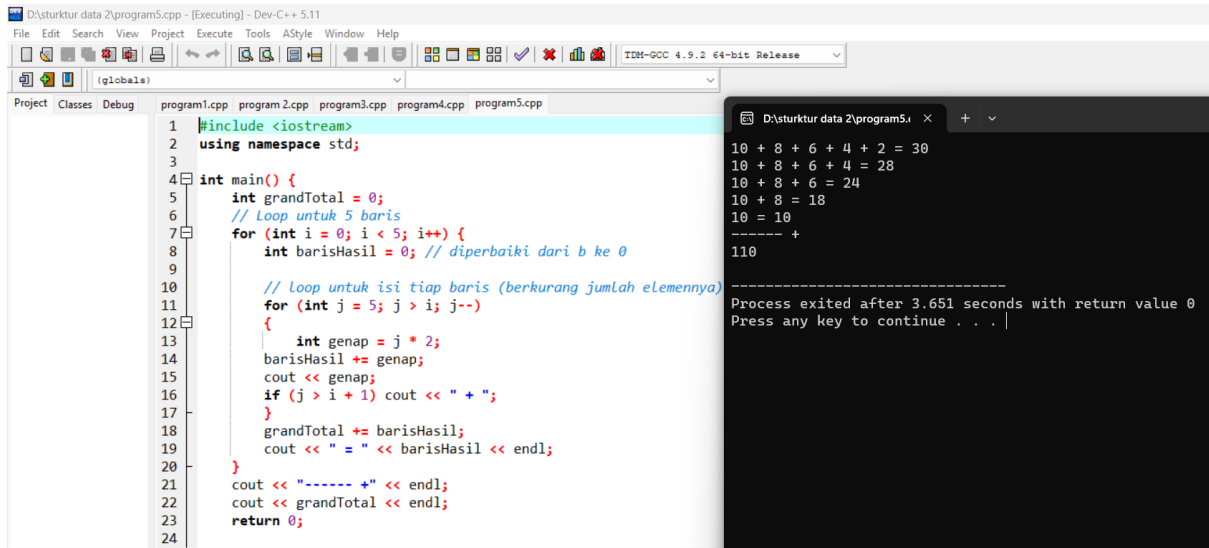
```
1 = 1
1 * 3 = 3
1 * 3 * 5 = 15
1 * 3 * 5 * 7 = 105
1 * 3 * 5 * 7 * 9 = 945

-----
Process exited after 1.558 seconds with return value 0
Press any key to continue . . .
```

5. Soal ke Lima

Buatlah program untuk menghitung perkalian deret bilangan genap membentuk segitiga siku terbalik dengan hasilnya :

JAWABAN:



The screenshot shows a C++ IDE with the source code for a program that calculates the product of even numbers in a reversed triangle. The code uses nested loops: an outer loop for the number of rows (5) and an inner loop for the number of even numbers in each row (decreasing from 5 to 1). The output window displays the resulting triangle of calculations and the final sum.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int grandTotal = 0;
6     // Loop untuk 5 baris
7     for (int i = 0; i < 5; i++) {
8         int barisHasil = 0; // diperbaiki dari b ke 0
9
10        // Loop untuk isi tiap baris (berkurang jumlah elemennya)
11        for (int j = 5; j > i; j--)
12        {
13            int genap = j * 2;
14            barisHasil += genap;
15            cout << genap;
16            if (j > i + 1) cout << " + ";
17        }
18        grandTotal += barisHasil;
19        cout << " = " << barisHasil << endl;
20    }
21    cout << "----- +" << endl;
22    cout << grandTotal << endl;
23    return 0;
24 }
```

Output:

```
10 + 8 + 6 + 4 + 2 = 30
10 + 8 + 6 + 4 = 28
10 + 8 + 6 = 24
10 + 8 = 18
10 = 10
----- +
110
Process exited after 3.651 seconds with return value 0
Press any key to continue . . .
```

Link Github: <https://github.com/syifaulaja3300/syifaulam-tm>